

Understanding Tina4

Architecture, Philosophy, and the Four-Language Promise

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The Intelligent Native Application 4ramework

Table of Contents

What Is Tina4? 7

The AI Framework Philosophy . . . 7

Why Zero Dependencies Matters . . . 8

Security . . . 8

Size . . . 8

Portability . . . 8

Upgrades . . . 8

The One Exception . . . 8

44 Features at 100% Parity . . . 9

Convention Over Configuration . . . 10

The Four-Language Paradigm . . . 11

What Tina4 Is Not . . . 11

Summary . . . 12

Architecture 13

The Request Lifecycle . . . 13

1. Server . . . 13

2. Request . . . 14

3. Router . . . 14

4. Middleware . . . 15

5. Handler . . . 15

6. Response . . . 16

7. Response Pipeline . . . 16

Project Structure . . . 16

src/routes/ -- Where Your API Lives . . . 17

src/orm/ -- Where Your Models Live . . . 17
src/migrations/ -- Database Schema Changes . . . 17
src/seeds/ -- Test Data . . . 18
src/templates/ -- Frond Templates . . . 18
src/public/ -- Static Files . . . 18
data/ -- Runtime Data . . . 18
logs/ -- Log Files . . . 18
secrets/ -- JWT Keys . . . 18
.env Driven Configuration . . . 18
The Priority Chain . . . 19
Example: Priority Chain in Practice . . . 19
is_truthy() -- Boolean Environment Values . . . 19
Dev Mode vs. Production Mode . . . 20
Dev Mode (TINA4_DEBUG=true) . . . 20
Production Mode (TINA4_DEBUG=false) . . . 20
The Frond Template Engine . . . 21
Basic Syntax . . . 21
How Auto-Discovery Works . . . 22
Routes . . . 22
Models . . . 23
Templates . . . 23
Static Files . . . 23
Summary . . . 24
Choosing Your Language 25
Python . . . 25
Why Choose Python . . . 25
Watch Out For . . . 25

Quick Start . . . 26
PHP . . . 26
Why Choose PHP . . . 26
Watch Out For . . . 27
Quick Start . . . 27
Ruby . . . 27
Why Choose Ruby . . . 27
Watch Out For . . . 28
Quick Start . . . 28
Node.js . . . 28
Why Choose Node.js . . . 28
Watch Out For . . . 29
Quick Start . . . 29
Comparison Table . . . 30
Switching Languages . . . 31
How to Decide . . . 31
Environment Variables 32
How .env Files Work . . . 32
Rules . . . 32
The .env File Is Not Committed to Git . . . 33
The Priority Chain . . . 33
Boolean Values . . . 33
Complete .env Reference . . . 34
Server . . . 34
Authentication . . . 35
CSRF Protection . . . 35
Database . . . 36

DB Query Cache . . . 37
ORM . . . 37
CORS (Cross-Origin Resource Sharing) . . . 37
Security Headers . . . 38
Rate Limiter . . . 38
Sessions . . . 39
Redis/Valkey Session Backend . . . 39
MongoDB Session Backend . . . 40
Queue . . . 40
RabbitMQ Queue Backend . . . 41
Kafka Queue Backend . . . 41
MongoDB Queue Backend . . . 41
Response Cache . . . 42
Logging . . . 43
Messenger (Email / SMTP) . . . 43
IMAP (for reading email) . . . 44
Localization (i18n) . . . 44
Swagger / OpenAPI . . . 44
File Uploads . . . 44
WebSocket . . . 45
Services (Background Workers) . . . 45
Dev Mailbox . . . 45
Minimal .env for Development . . . 45
Minimal .env for Production . . . 46
Docker .env . . . 46
Full .env Template . . . 46
Summary . . . 49

DevOps and Deployment 50

1. Know Which Server You Are Running . . . 50

2. PHP Only: Pick a Runtime . . . 50

php-fpm is the safe default . . . 51

The built-in server needs pcntl . . . 51

Forking is the default, so there is nothing to turn on . . . 51

Ask the server, do not trust the config . . . 52

Swoole keeps your application resident . . . 52

The Swoole entry point . . . 52

3. Node.js and Ruby: What Each One Does With a Slow Handler . . . 54

4. Environment Variables That Matter in Production . . . 55

Set these or the app tells you off . . . 55

Timeouts that stop a hang . . . 55

Limits that bound a hostile request, PHP only . . . 55

Worker pool, PHP built-in server only . . . 56

5. Health Checks . . . 56

6. Graceful Shutdown . . . 56

7. Logging . . . 57

8. A Checklist . . . 57

What Is Tina4?

The AI Framework Philosophy

TINA4: The Intelligent Native Application 4ramework.

Four words. Each one carries weight.

Intelligent. The framework understands AI. Every project ships with a CLAUDE.md file that gives AI assistants complete knowledge of the framework's conventions, API, and structure. Your AI writes correct code on the first try because the framework told it how.

Native. Zero third-party dependencies. Every feature (the template engine, the JWT library, the SCSS compiler, the queue system, the GraphQL parser) is built from scratch using the language's standard library. No supply chain risk. No version conflicts. No surprises.

Application. This builds real production applications. Routing, ORM, authentication, queues, WebSocket, email, GraphQL, SOAP, 44 features, all included. One package. One install. Ship today.

4ramework. Four languages. One API. Python, PHP, Ruby, Node.js: learn the conventions once, build in any of them. The "4" is both the number and "for", a framework *for* developers who value their time.

Here is a complete API endpoint:

```
<?php
// src/routes/greeting.php

Router::get("/api/greeting/{name}", function ($request, $response) {
    return $response->json([
        "message" => "Hello, " . $request->params["name"]
    ]);
});
```

No base controller. No service provider. No bootstrapping ritual. Drop that file into `src/routes/`, start the server, and it works. Your AI assistant knows this too: it reads the same conventions you do.

The philosophy fits in one sentence: **the framework that gets out of the way, for humans AND for AI.**

Routes go in `src/routes/`. Templates go in `src/templates/`. Models go in `src/orm/`. Learn the convention once. Your AI learns it once. Neither of you thinks about it again.

A decade of watching developers waste afternoons on configuration files, dependency conflicts, and framework upgrades that break everything. Then AI arrived and made the problem worse: every framework's ambiguity became the AI's confusion. Tina4 was born from both frustrations. One structure. One way to do things. The AI never guesses wrong because there is only one right answer.

Why Zero Dependencies Matters

Tina4 v3 has **zero third-party dependencies** for its core features. The template engine, the JWT library, the SCSS compiler, the queue system, the GraphQL parser, the logger, the rate limiter: every piece is built from scratch using the language's standard library.

This is a survival strategy, for you and for your AI assistant.

Security

Every dependency is an attack surface. When a package in your dependency tree gets compromised (and it will, ask the teams who trusted `event-stream`, `colors.js`, or `ua-parser-js`) your application is exposed.

Tina4's attack surface is the language runtime and your code. Nothing else sits between you and your users.

Size

A Laravel installation pulls in 70+ packages. A Rails app starts with 40+ gems. A Next.js project's `node_modules` folder is measured in hundreds of megabytes.

Tina4 installs **one package**. The framework runs to roughly **~26,000 lines of code** per language (Python ~26,000 | PHP ~35,000 | Ruby ~24,000 | Node.js ~32,000), all of it standard-library code you can read and audit. The Docker image fits in **40-80MB**. Your production container ships with what it needs. Nothing else tags along.

Portability

Zero dependencies means zero compatibility conflicts. You will never see this with Tina4:

```
Your requirements could not be resolved to an installable set of packages.
  Problem 1
    - package-a v2.1 requires other-package ^3.0
    - package-b v1.4 requires other-package ^2.0
```

No diamond dependency problem. No dependency tree to untangle. No Friday afternoon emergency because a transitive dependency released a breaking change.

Upgrades

Upgrading Tina4 means upgrading one package. No cascade of breaking changes. The framework team controls every line, so when something breaks, the fix lives in one place.

The One Exception

Database drivers are the exception. You cannot talk to PostgreSQL without a PostgreSQL driver. These are native connectors to external systems. They are optional: install only what you need. SQLite works out of the box with every language's standard library.

---

44 Features at 100% Parity

Tina4 ships with everything you need to build a production web application. 44 features, all implemented identically across Python, PHP, Ruby, and Node.js. 9,311 tests across 280 test files. Here is what every installation includes:

Core Web

- HTTP router with path parameters, typed params, middleware, and auth guards
- Request and Response objects with full HTTP access
- Static file serving, CORS, rate limiting, health checks
- Graceful shutdown, request ID tracking, structured logging
- Response compression, ETag support

Data Layer

- SQL-first ORM with Active Record pattern
- Five database drivers: SQLite, PostgreSQL, MySQL, MSSQL, Firebird
- Relationships: hasOne, hasMany, belongsTo with eager loading
- Migrations with rollback, seeders with 50+ fake data generators
- Query result caching with TTL, paginated results

Template and Frontend

- Frond: a Twig-compatible template engine with 55+ filters
- Template inheritance, includes, macros, pre-compilation
- SCSS compiler, tina4css (built-in CSS framework), frond.js (frontend helpers)

Auth and Sessions

- JWT (HS256/RS256) built from scratch
- Four session backends: file, Redis, Valkey, MongoDB
- CSRF protection, password hashing

Integration

- Queue system with retry, dead letters, and four backends (SQLite, RabbitMQ, Kafka, MongoDB)
- GraphQL parser and executor
- WebSocket server
- SOAP/WSDL support, HTTP API client, email messenger, i18n

Developer Experience

- Rust-based unified CLI with scaffolding, migrations, and testing
- Dev admin dashboard with 11 panels

- Error overlay with source code and stack traces
- Interactive gallery with 7 deployable examples
- Live reload, AI tool integration

Each language packages these capabilities in its native format. A dependency means an extra package that the package manager installs; a language or runtime extension does not count. Python and PHP core declare no required third-party packages. Ruby declares runtime gems, while Node's root package installs optional provider packages by default. The biggest component, the Frond template engine, runs about 1,500 lines. Most features need fewer than 200.

Convention Over Configuration

Tina4 projects follow a predictable structure. Run `tina4 init` and you get:

```
my-project/
|-- .env                    # Configuration
|-- src/
|   |-- routes/             # Route handlers (auto-discovered)
|   |-- orm/                # ORM models (auto-discovered)
|   |-- templates/          # Frond templates
|   |-- public/             # Static files (served directly)
|   |   |-- css/
|   |   |-- js/
|   |   `-- images/
|   `-- scss/               # SCSS source files (auto-compiled)
|-- migrations/             # SQL migration files
|-- data/                   # SQLite databases (gitignored)
|-- logs/                   # Log files with rotation (gitignored)
`-- tests/                  # Test files
```

Five rules. No exceptions:

- **Routes** go in `src/routes/`. Name the files however you want. Tina4 reads the route definitions inside them.
- **Models** go in `src/orm/`. Same auto-discovery.
- **Templates** go in `src/templates/`. Call `response.render("products/list.twig", data)` and Tina4 finds it.
- **Static files** go in `src/public/`. A file at `src/public/css/style.css` serves at `/css/style.css`.
- **Configuration** goes in `.env`. One file. Key-value pairs. No YAML. No TOML. No JSON config.

No routing table to maintain. No service container to wire up. No middleware stack to arrange in the right order. Drop files in the right directories. They work.

The Four-Language Paradigm

Tina4 is one framework with four backend implementations:

- **tina4-python** - Python 3.12+
- **tina4-php** - PHP 8.2+
- **tina4-ruby** - Ruby 3.1+
- **tina4-nodejs** - Node.js 22+ (TypeScript)

All four target the same project structure, `.env` variables, template syntax, CLI commands, and API contracts. The audit records gaps until fixtures prove those contracts in every implementation.

Each implementation follows its language's naming convention:

Concept	Python / Ruby	PHP / Node.js
Method names	`snake_case`	`camelCase`
Fetch one row	`fetch_one()`	`fetchOne()`
Soft delete	`soft_delete()`	`softDelete()`

A team can prototype in Python and deploy in PHP without relearning the framework. Frontend developers using frond.js never need to know which backend language is running. DevOps deploys the same Docker structure, the same `.env`, the same health checks, regardless of language.

One Rust-based CLI binary auto-detects the project language and dispatches to the correct runtime:

```
tina4 init python ./my-app    # Scaffold a Python project
tina4 serve                   # Start dev server
tina4 generate model User     # Generate an ORM model
tina4 migrate                 # Run pending migrations
tina4 test                    # Run the test suite
```

What Tina4 Is Not

Tina4 does not replace Laravel, Django, Rails, or Next.js. Those are excellent frameworks for teams that want a full-stack opinion on everything.

Tina4 is for developers who want:

- **Control** - you see every line of code that runs your application
- **Simplicity** - one package, one import, predictable behaviour
- **Speed** - sub-millisecond framework overhead
- **Portability** - switch languages without switching paradigms
- **Security** - no supply chain risk from transitive dependencies

If you want a batteries-included platform with an ecosystem of plugins and a marketplace of themes, Tina4 is the wrong tool. If you want a sharp, minimal toolkit that does what you tell it and nothing else, keep reading.

The code you don't write is the code that never breaks.

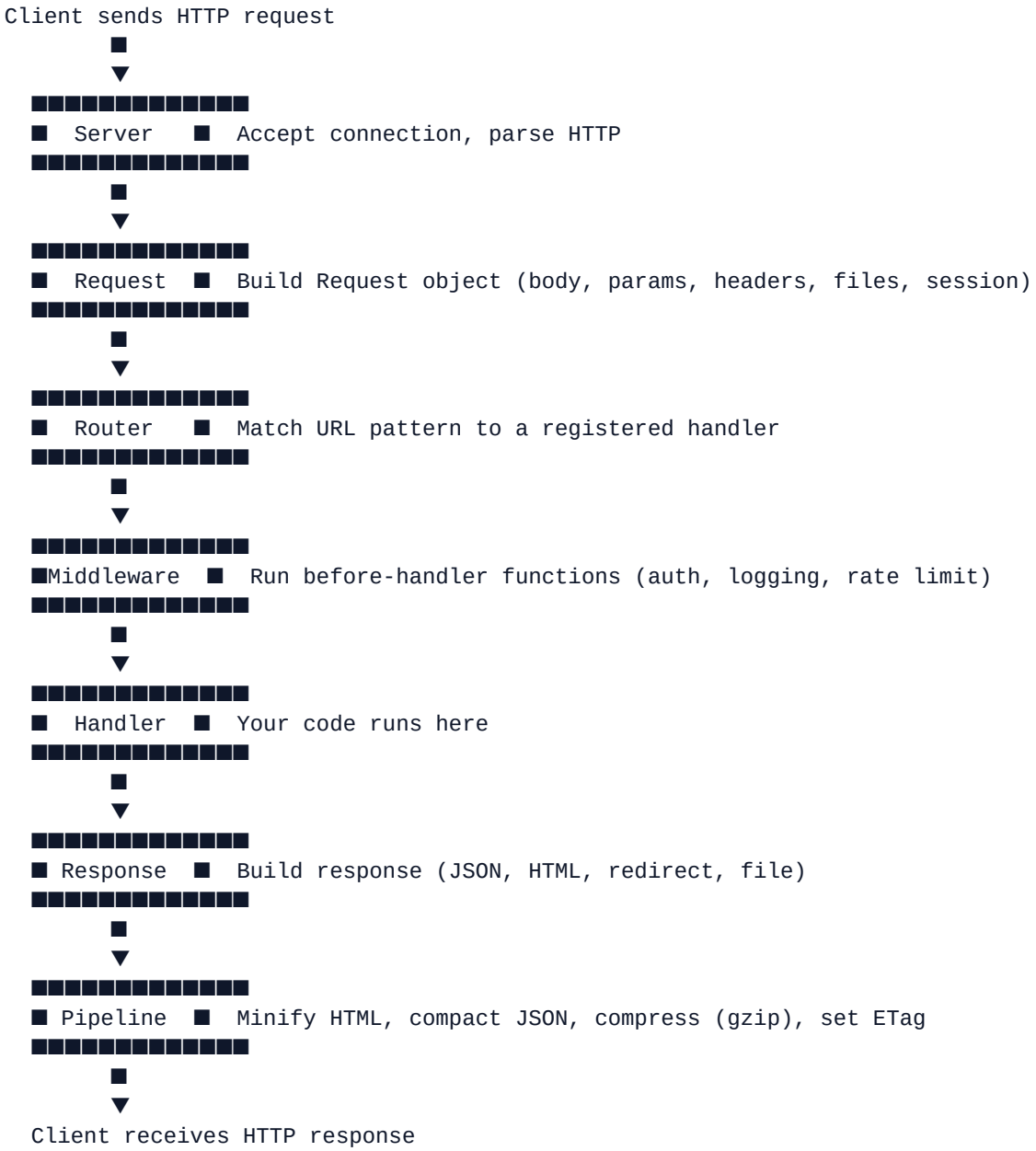
Summary

Aspect	Tina4
Philosophy	Toolkit, not a cathedral
Dependencies	Zero (core features)
Framework size	~26,000 lines per language (avg)
Languages	Python, PHP, Ruby, Node.js
Configuration	`.env` file only
Discovery	Automatic (routes, models, templates)
CLI	Unified Rust binary
Tests	9,311 across all four frameworks
Features	44 at 100% parity

Architecture

The Request Lifecycle

A request arrives. Seven stages later, a response leaves. Every Tina4 application follows this path -- Python, PHP, Ruby, Node.js. The language changes. The architecture does not.



Learn this diagram. Everything else in Tina4 is a footnote to it.

1. Server

The server accepts TCP connections and parses raw HTTP into structured data. Each language uses its native server:

Language	Server

Python	asyncio / ASGI
PHP	Built-in server / Swoole
Ruby	WEBrick / Puma
Node.js	`node:http`

You never touch this layer. Tina4 owns it. The server starts, listens, and hands off parsed requests. Your code lives further down the chain.

2. Request

The framework assembles a `Request` object. Everything the handler needs is already unpacked and waiting:

```
request.body        # Parsed JSON or form data
request.params      # Path parameters ({id} from /users/{id})
request.query       # Query string (?page=2&sort=name)
request.headers     # HTTP headers (case-insensitive)
request.files       # Uploaded files
request.session     # Session data (read/write)
request.method      # GET, POST, PUT, DELETE, etc.
request.path        # URL path without query string
request.ip          # Client IP address
request.request_id  # Unique ID for this request (for log correlation)
request.cookies     # Parsed cookies
request.is_json     # True if Content-Type contains "json"
```

No parsing. No extraction. No guessing where the data lives. One object. Twelve properties. Everything accounted for.

3. Router

The router listens. A request arrives. Method and path are matched against registered routes. Routes live in files under `src/routes/`:

```
// PHP
Router::get("/api/products/{id:int}", function ($request, $response) {
    // Only matches if {id} is an integer
});
```

```
# Python
@get("/api/products/{id:int}")
async def get_product(request, response):
    # Only matches if {id} is an integer
    pass
```

```
# Ruby
get "/api/products/{id:int}" do |request, response|
  # Only matches if {id} is an integer
end
```

```
// Node.js (file-based routing: src/routes/api/products/[id].ts)
export default function handler(request, response) {
    // id available via request.params.id
}
```

The router supports:

- **Basic parameters:** `/users/{id}` matches `/users/42`
- **Typed parameters:** `/users/{id:int}` only matches integers
- **Catch-all:** `/pages/{slug:.*}` matches `/pages/about/team/history`
- **Route groups:** Prefix multiple routes with a common path and middleware
- **Route caching:** Cache the response for a given TTL

No match on a registered route? The router checks `src/public/` for a static file. Still nothing? A 404 goes back to the client.

4. Middleware

Middleware functions form a pipeline. Each one inspects the request, decides whether to pass it forward, and optionally modifies what comes back. Think of it as a series of gates. A request must pass through every gate before it reaches your handler.

Tina4 ships four built-in middleware:

- **CORS** -- configured via environment variables, runs on every request
- **Rate limiting** -- 60 requests per minute per IP, out of the box
- **Auth gating** -- attach `.secure()` to a route to demand a valid JWT
- **Request ID tracking** -- generates or reads the `X-Request-ID` header

You write your own the same way:

```
// PHP
function logRequests($request, $response, $next) {
    $start = microtime(true);
    $result = $next($request, $response);
    $duration = round((microtime(true) - $start) * 1000, 2);
    Log::info("Request completed", [
        "method" => $request->method,
        "path" => $request->path,
        "duration_ms" => $duration
    ]);
    return $result;
}

Router::get("/api/users", $handler)->middleware([logRequests]);
```

A middleware receives the request, calls `$next` to continue the chain, and returns the result. Skip the `$next` call and the request stops right there. Short-circuit. The handler never runs. This is how auth guards work -- no valid token, no entry.

5. Handler

This is your territory. The handler receives a `Request` and a `Response`. What happens in between is your decision. Tina4 does not impose an application architecture. No base controllers. No service containers. No required inheritance. You receive two objects. You return a response.

6. Response

The `Response` object covers every common output:

```
response.json(data, statusCode)       # JSON with auto Content-Type
response.html(content, statusCode)    # HTML response
response.text(content, statusCode)    # Plain text
response.xml(content, statusCode)     # XML response
response.redirect(url, statusCode)    # HTTP redirect (302 or 301)
response.file(path)                   # File download with auto MIME type
response.render(template, data)       # Render a Frond template
response.status(code)                 # Set status code (chainable)
response.header(name, value)          # Set response header (chainable)
response.cookie(name, value, options) # Set a cookie
```

Ten methods. JSON, HTML, text, XML, redirects, files, templates, status codes, headers, cookies. Pick the one that fits. Chain what needs chaining.

7. Response Pipeline

Your handler finishes. The response is not done yet. It passes through an automatic pipeline -- five stages that optimize every response without a single line of configuration:

- **Frond rendering** -- if you called `response.render()`, the template compiles and executes
- **HTML minification** -- in production (`TINA4_DEBUG=false`), whitespace collapses, comments vanish. 15-25% smaller output.
- **JSON compaction** -- JSON ships compact. Add `?pretty=true` to the query string during development for readable output.
- **gzip compression** -- the client sends `Accept-Encoding: gzip` and the response exceeds 1KB? Compressed.
- **ETag generation** -- a hash of the response body becomes an `ETag` header. The next request with a matching `If-None-Match` gets a `304 Not Modified`. Zero bytes transferred.

Five optimizations. Zero configuration. Every response benefits.

Project Structure

Every Tina4 project follows the same directory layout. Python, PHP, Ruby, Node.js -- the folders are identical. A developer who has seen one Tina4 project has seen them all.

```
my-project/
■■■ .env                    # All configuration lives here
■■■ src/
■   ■■■ routes/             # Route handlers (auto-discovered)
■   ■■■ orm/                # ORM models (auto-discovered)
■   ■■■ migrations/         # SQL migration files
■   ■■■ seeds/              # Database seed files
■   ■■■ templates/          # Frond templates
■   ■   ■■■ errors/         # Custom error pages
■   ■■■ public/             # Static files (served at /)
■   ■   ■■■ js/
```

```
■   ■   ■   ■■■ frond.js    # Auto-provided
■   ■   ■■■ css/
■   ■   ■■■ scss/           # SCSS source files (auto-compiled)
■   ■   ■■■ images/
■   ■   ■■■ icons/
■   ■■■ locales/            # Translation files (JSON)
■       ■■■ en.json
■■■ data/                  # SQLite databases, .broken files
■■■ logs/                  # Log files with rotation
■■■ secrets/               # JWT keys
■■■ tests/                 # Test files
```

Fourteen directories. Each one has a single purpose. No overlap. No ambiguity.

`src/routes/` -- Where Your API Lives

Drop a file here. Tina4 finds the route definitions inside it. Organize however you want:

```
src/routes/
■■■ products.php        # All product routes
■■■ orders.php          # All order routes
■■■ admin/
    ■■■ users.php       # Admin user routes
    ■■■ reports.php     # Admin report routes
```

One file or twenty. Nested folders or flat. Tina4 reads them all. The file name does not affect the route path. Only the route definition inside the file matters. Name it `products.php` or `banana.php` -- the URL comes from the code, not the filename.

`src/orm/` -- Where Your Models Live

ORM model classes go here. Auto-discovered on startup:

```
src/orm/
■■■ Product.php
■■■ Order.php
■■■ OrderItem.php
■■■ User.php
```

Drop a class that extends the ORM base. Tina4 registers it. Auto-CRUD endpoints, route model binding, relationship resolution -- all of it flows from discovery.

`src/migrations/` -- Database Schema Changes

Migrations are SQL files with timestamps:

```
src/migrations/
■■■ 20260319100000_create_users_table.sql
■■■ 20260319100000_create_users_table.down.sql
■■■ 20260320090000_create_products_table.sql
■■■ 20260320090000_create_products_table.down.sql
```

The `.sql` file runs on `tina4 migrate`. The `.down.sql` file runs on `tina4 migrate --rollback`. Tina4 tracks which migrations have run in a `tina4_migrations` table. Forward and back. Always reversible.

`src/seeds/` -- Test Data

Seed files populate your database with test or default data. Run them with `tina4 seed`. Fifty built-in fake data generators handle the rest -- names, emails, addresses, phone numbers, dates.

`src/templates/` -- Frond Templates

Templates use the Frond engine. Inheritance, includes, filters, loops, conditionals -- all covered in detail later. The structure is yours to decide:

```
src/templates/
■■■ base.html           # Layout with blocks
■■■ index.html          # Extends base.html
■■■ products/
■   ■■■ list.html       # Product listing
■   ■■■ detail.html     # Single product
■■■ partials/
■   ■■■ header.html
■   ■■■ footer.html
■■■ errors/
    ■■■ 404.html        # Custom 404 page
    ■■■ 500.html        # Custom 500 page
```

`src/public/` -- Static Files

Files here are served at the root URL path. A file at `src/public/images/logo.png` appears at `/images/logo.png`. No route registration. No configuration. Drop it in. It serves.

The framework auto-provides `frond.js` in `src/public/js/` and keeps it in sync with the installed framework version. You never manage this file.

`data/` -- Runtime Data

SQLite databases live here by default (`data/app.db`). The `.broken/` subdirectory holds error marker files used by the health check. This entire directory belongs in `.gitignore`. Runtime data stays on the machine that runs the application.

`logs/` -- Log Files

Structured log files with automatic rotation. In `.gitignore`. Compressed after two days. Deleted after thirty. The framework handles all of it.

`secrets/` -- JWT Keys

Private and public keys for RS256 JWT signing. In `.gitignore`. Generated once, deployed with your application, never committed to source control.

.env Driven Configuration

One file controls everything. Not YAML. Not TOML. Not JSON config objects. A `.env` file at the project root. Key-value pairs. Plain text.

```
# .env
TINA4_DEBUG=true
```

```
TINA4_PORT=7145
TINA4_DATABASE_URL=sqlite:///data/app.db
JWT_SECRET=change-me-in-production
```

Four lines. A working application.

The Priority Chain

Tina4 resolves every configuration value through a three-level chain:

```
Constructor argument  >  .env file  >  Hardcoded default
```

The constructor wins. Always. The `.env` file is second. The hardcoded default is the safety net. This pattern applies to every configurable value in the framework. No exceptions.

Example: Priority Chain in Practice

```
# .env
TINA4_PORT=8080
```

```
// In code -- this overrides .env
$app = new Tina4\App(["port" => 9000]);
// Server starts on port 9000, not 8080
```

```
// No code override, no .env value
$app = new Tina4\App();
// Server starts on port 7145 (the default)
```

Three levels. Predictable resolution. Every time.

is_truthy() -- Boolean Environment Values

Environment variables are strings. Booleans do not exist in `.env` files. Tina4 bridges this gap with `is_truthy()`. Four values mean `true`:

- `true` (any case: `True`, `TRUE`, `tRuE`)
- `1`
- `yes` (any case)
- `on` (any case)

Everything else is `false`. Empty strings. Unset variables. Typos. If it is not on the list, it is `false`.

```
TINA4_DEBUG=true
TINA4_DEBUG=True
TINA4_DEBUG=1
TINA4_DEBUG=yes
TINA4_DEBUG=on
```

All equivalent. All enable debug mode.

```
TINA4_DEBUG=false
TINA4_DEBUG=0
TINA4_DEBUG=no
TINA4_DEBUG=off
TINA4_DEBUG=
# or simply omit the line
```

All disable it. No ambiguity.

Dev Mode vs. Production Mode

One variable. Two personalities. `TINA4_DEBUG` controls everything.

Dev Mode (`TINA4_DEBUG=true`)

The framework opens up. Every diagnostic tool activates:

- **Debug overlay** injects into every HTML response -- a toolbar showing request details, database queries, template render times, session data, and logs
- **Full stack traces** appear in the browser with source code context, the triggering request, and the queries that ran
- **Swagger UI** auto-registers at `/swagger`
- **Admin console** becomes available at `/__dev`
- **Live reload** watches for file changes and refreshes the browser
- **SQL query logging** writes every query to `logs/query.log`
- **Pretty JSON** is available via `?pretty=true` on any JSON endpoint
- **404 pages** show helpful route-not-found messages listing similar registered routes

Development mode assumes you want to see everything. It shows you everything.

Production Mode (`TINA4_DEBUG=false`)

The framework locks down. Every diagnostic tool disappears:

- **No debug overlay** -- responses ship clean
- **Generic error pages** -- no stack traces, no source code, no query details reach the browser
- **HTML minification** -- comments stripped, whitespace collapsed, 15-25% smaller output
- **.broken files** -- unhandled exceptions create marker files in `data/.broken/` that flip the health check to `503 Service Unavailable`, triggering container restarts
- **No Swagger UI** -- unless you force it on
- **No dev dashboard** -- only available when `TINA4_DEBUG=true`
- **No query logging** -- unless you enable it
- **Compact JSON only** -- no `?pretty=true`

Production mode assumes you want to expose nothing. It exposes nothing.

The default is `TINA4_DEBUG=false`. Forget to set it? The safe thing happens. Your application starts locked down.

One mistake will undo all of this: deploying with `TINA4_DEBUG=true`. Stack traces, database queries, session data -- visible to anyone with a browser. Set `TINA4_DEBUG=false` in production. Always.

The Frond Template Engine

Frond is Tina4's template engine. Zero dependencies. Built from scratch in each language. The syntax borrows from Twig -- developers who know Twig, Jinja2, or Nunjucks will recognize every construct. But no third-party template library runs underneath. Frond is Tina4's own.

Basic Syntax

Variables:

```
<h1>{{ title }}</h1>
<p>Welcome, {{ user.name }}</p>
<p>First item: {{ items[0] }}</p>
```

Filters (pipe syntax):

```
<p>{{ name | upper }}</p>           <!-- JOHN DOE -->
<p>{{ name | lower }}</p>           <!-- john doe -->
<p>{{ price | number_format(2) }}</p> <!-- 29.99 -->
<p>{{ text | truncate(100) }}</p>    <!-- First 100 chars... -->
<p>{{ description | raw }}</p>       <!-- No auto-escaping -->
<p>{{ items | length }}</p>          <!-- 5 -->
<p>{{ items | join(", ") }}</p>      <!-- apple, banana, cherry -->
```

Fifty-five filters. Strings, numbers, dates, arrays, encoding, formatting. If you need to transform data in a template, a filter exists.

Control structures:

```
{% if products | length > 0 %}
    {% for product in products %}
        <div class="product">
            <h2>{{ product.name }}</h2>
            <p>{{ product.price | number_format(2) }}</p>
            {% if loop.last %}
                <hr>
            {% endif %}
        </div>
    {% else %}
        <p>No products found.</p>
    {% endfor %}
{% endif %}
```

Template inheritance:

```
{# base.html #}
<!DOCTYPE html>
<html>
<head>
    <title>{% block title %}My App{% endblock %}</title>
</head>
<body>
    {% block content %}{% endblock %}
</body>
</html>

{# products/list.html #}
{% extends "base.html" %}
```

```
{% block title %}Products{% endblock %}

{% block content %}
    <h1>Our Products</h1>
    {% for product in products %}
        <p>{{ product.name }} - ${{ product.price | number_format(2) }}</p>
    {% endfor %}
{% endblock %}
```

Includes:

```
{% include "partials/header.html" %}
<main>
    {{ content }}
</main>
{% include "partials/footer.html" %}
```

Fragment caching:

```
{% cache "product-list" 300 %}
    {# This block is cached for 300 seconds #}
    {% for product in products %}
        <div>{{ product.name }}</div>
    {% endfor %}
{% endcache %}
```

One fact matters above all others: the template syntax is identical across all four languages. A template written for a Python backend works on PHP, Ruby, and Node.js without a single change. The backend is invisible to the frontend. Frond guarantees that.

How Auto-Discovery Works

Tina4 finds your code without being told where to look. No registration files. No import chains. No bootstrapping rituals. Drop files in the right directories. The framework discovers them.

Routes

On startup, Tina4 scans every file in `src/routes/` -- recursively, through every subdirectory. It finds route registration calls: `get()`, `post()`, `put()`, `delete()`, `any()`. Each call registers a route with the router.

```
Startup
  ■
  ■■■ Scan src/routes/
  ■   ■■■ products.php → registers GET /api/products, POST /api/products, ...
  ■   ■■■ orders.php → registers GET /api/orders, ...
  ■   ■■■ admin/users.php → registers GET /api/admin/users, ...
  ■
  ■■■ Router now has all routes in memory
```

The scan happens once at startup. In dev mode with live reload, the scan re-runs when files change. New route file saved? The router rebuilds. No restart needed.

The file name and path within `src/routes/` do not determine the URL. Only the route definition inside the file matters. Put all your routes in one file called `everything.php` and it

works. Spread them across fifty files in nested folders and it works. The directory structure is for your organization. The framework ignores it.

Models

ORM model classes in `src/orm/` follow the same pattern. The framework scans for classes that extend the ORM base class and registers them. This powers auto-CRUD -- REST endpoints generated from models. It powers route model binding -- URL parameters resolved to model instances. Discovery makes both possible without a single line of registration code.

Templates

Templates work differently. They are not discovered at startup -- they are loaded on demand when `response.render()` is called. But the framework knows where to look without being told. Reference `"products/list.html"` and Tina4 resolves it to `src/templates/products/list.html`. No path configuration. No template registry.

Static Files

When the router cannot match a request to a registered route, it falls back to the filesystem. The lookup follows a strict order:

```
1. Registered route?          → Run handler
2. File in src/public/?       → Serve static file
3. Framework built-in asset?  → Serve framework file
4. Nothing matches            → 404 response
```

Four steps. Tried in order. The first match wins. A file at `src/public/css/style.css` serves at `/css/style.css` with the correct MIME type. No route needed. No configuration needed. The file exists, so it serves.

Summary

Concept	How It Works in Tina4
Request lifecycle	Request > Router > Middleware > Handler > Response > Pipeline
Project structure	Fixed conventions: routes, orm, templates, public, migrations
Configuration	`.env` only, priority: constructor > .env > default
Dev vs. production	Single toggle: `TINA4_DEBUG`
Template engine	Frond: Twig-like syntax, zero dependencies, identical across languages
Auto-discovery	Files in `src/routes/` and `src/orm/` are found at startup
Static files	Files in `src/public/` are served at `/`
Response pipeline	Minification > compression > ETag, all automatic

Choosing Your Language

Tina4 runs on four languages. Same API. Same project structure. Same template syntax. Same CLI commands. The framework disappears -- what remains is your team, your hosting, and your problem.

This chapter gives you the information to choose.

Python

Best for: Data science teams, ML/AI integration, async-heavy applications, rapid prototyping.

```
# src/routes/products.py
from tina4_python import get, post

@get("/api/products")
async def list_products(request, response):
    products = Product.select(page=request.query.get("page", 1))
    return response.json(products)

@post("/api/products")
async def create_product(request, response):
    product = Product.create(request.body)
    return response.json(product, 201)
```

Why Choose Python

Async-native. Tina4 Python runs on asyncio. Every route handler is an async function. External API calls, database queries, file processing -- async means your server handles thousands of concurrent connections without spawning threads. One event loop. Thousands of requests.

Data science integration. Your web application sits next to a machine learning pipeline. Your API endpoint imports pandas, numpy, or scikit-learn directly. No inter-process communication. No REST calls to a separate service. The model runs in the same process as the route handler. Train in a notebook, deploy behind an endpoint.

The reference implementation. Tina4 Python was the first v3 framework to reach 100% completion. When ambiguity exists in the spec, Python's behavior is canonical. Every other implementation follows its lead.

Largest general-purpose community. Python is the most-taught programming language in universities worldwide. Finding Python developers is straightforward -- new graduates learn it first, data scientists think in it, and the library set grows faster than any other language.

Watch Out For

GIL limitations. CPU-bound work is single-threaded. The Global Interpreter Lock prevents true parallel execution of Python code. Image processing, PDF generation, heavy computation -- offload these to background tasks or a queue worker process. The event loop handles I/O. The GIL handles everything else, one thread at a time.

Deployment overhead. Python deployment demands more setup than PHP. Virtual environments, pip dependencies (Tina4 has zero, but your application might not), process managers. Docker simplifies this. Without Docker, plan for systemd units and environment isolation.

Package naming. Install with `pip install tina4-python`. The CLI command is `tina4` (auto-detects language). The import is `from tina4_python import ...`. Three different names for the same thing. Learn them once.

Quick Start

```
pip install tina4-python
tina4 init python my-project
cd my-project
tina4 serve
```

PHP

Best for: Existing PHP teams, shared hosting, CMS-adjacent projects, the widest hosting support on the planet.

```
<?php
// src/routes/products.php
use Tina4Router;

Router::get("/api/products", function ($request, $response) {
    $products = Product::select(page: $request->query["page"] ?? 1);
    return $response->json($products);
});

Router::post("/api/products", function ($request, $response) {
    $product = Product::create($request->body);
    return $response->json($product, 201);
});
```

Why Choose PHP

Hosting is everywhere. Shared hosting, VPS, dedicated servers, cloud platforms -- if it has a web server, it runs PHP. Every major hosting provider supports it. Every $5/month shared plan includes it. When your client's infrastructure is not negotiable, PHP fits.

Fastest Tina4. PHP 8.1+ with JIT compilation makes Tina4 PHP the fastest of the four implementations in raw request throughput. Add OPcache (compiled bytecode caching) and the framework overhead approaches zero. Requests arrive. Responses leave. The interpreter barely warms up.

Thirty years of libraries. Database drivers, payment processing, PDF generation, image manipulation -- PHP has well-tested libraries for all of it. Tina4 itself has zero dependencies. Your application can still pull in Composer packages when the need is real.

Monorepo simplicity. Tina4 PHP v3 is a single Composer package under the `Tina4\` namespace. No split packages. No sub-repositories. One `composer require`. Everything arrives.

Familiar ground. PHP was the first server-side language for a generation of web developers. If your team writes PHP, Tina4 speaks their language. The learning curve is the framework, not the language.

Watch Out For

No native async. Standard PHP is synchronous. One request, one thread, start to finish. For WebSocket support and true async behavior, you need the Swoole or OpenSwoole extension. Most web applications never need this. But if you plan to hold thousands of persistent connections, verify that Swoole is available on your hosting platform before you commit.

Extension management. PHP database drivers are C extensions: ext-pgsql, ext-mysqli, ext-sqlite3. On some hosting platforms, enabling these requires a support ticket or a PHP recompilation. Check your target platform's available extensions before you choose your database.

Case sensitivity. PHP uses `camelCase` for methods: `fetchOne()`, `softDelete()`, `hasMany()`. Coming from Python or Ruby? This is a style adjustment. The API is identical in capability -- the casing is different.

Quick Start

```
composer require tina4stack/tina4php
tina4 init php my-project
cd my-project
tina4 serve
```

Ruby

Best for: Startups, elegant code, Rails refugees who want less framework and more control.

```
# src/routes/products.rb
get "/api/products" do |request, response|
  products = Product.select(page: request.query["page"] || 1)
  response.json(products)
end

post "/api/products" do |request, response|
  product = Product.create(request.body)
  response.json(product, 201)
end
```

Why Choose Ruby

Elegant syntax. Ruby was designed to make programmers happy. That philosophy shows. Route definitions read as English. Blocks feel natural. The DSL is clean. Code written in Tina4 Ruby looks the way you think about it.

Rails without the weight. You built Rails applications. You found yourself fighting the framework more than using it. Tina4 Ruby gives you the parts you valued -- convention over configuration, migrations, ORM, template engine -- and drops the parts you did not. No massive dependency tree. No opaque internals. No 15-minute boot time on a large codebase.

Startup velocity. Ruby's expressiveness means less code for the same functionality. Fewer lines. Fewer files. Faster iteration. For teams that need to ship and adjust, Ruby shrinks the distance between idea and production.

Strong testing culture. The Ruby community treats testing as a first-class concern. Tina4 Ruby integrates with Ruby's testing tools. The framework itself carries 2,333 tests. Testing is not an afterthought. It is the foundation.

Watch Out For

Smaller hosting pool. Ruby hosting is less common than PHP. Shared hosting with Ruby support is rare. Plan on a VPS, container platform, or PaaS -- Heroku, Render, Fly.io. The options exist. They are just not everywhere.

Performance. Ruby is not the fastest language. For raw throughput, PHP and Node.js outperform it. For most web applications, Ruby is fast enough. But if you are building a high-traffic API that must serve thousands of requests per second on a single instance, benchmark with your actual workload first. Do not assume.

Smaller talent pool. Ruby developers are harder to find than Python or JavaScript developers. The community is passionate but compact. If you are scaling a team, factor hiring timelines into the decision.

Quick Start

```
gem install tina4ruby
tina4 init ruby my-project
cd my-project
tina4 serve
```

Node.js

Best for: JavaScript/TypeScript teams, file-based routing, real-time applications, highest raw throughput.

```
// src/routes/api/products/index.ts
export default function handler(request, response) {
    const products = Product.select({ page: request.query.page || 1 });
    return response.json(products);
}

// src/routes/api/products/index.post.ts
export default function handler(request, response) {
    const product = Product.create(request.body);
    return response.json(product, 201);
}
```

Why Choose Node.js

One language everywhere. Frontend and backend. Same developers. Same language. Same tooling. No context switching. A React component and its API endpoint live in the same mental model. JavaScript runs on both sides of the wire.

File-based routing. Tina4 Node.js maps file paths to URL paths. A file at `src/routes/api/products/[id].ts` handles `/api/products/42`. The filesystem is the routing table. If you have used Next.js, this pattern is already muscle memory.

Highest raw throughput. Node.js on V8 handles the most requests per second of the four implementations. For APIs that serve JSON and move data, Node.js delivers the highest ceiling. The event loop is purpose-built for I/O.

TypeScript from the ground up. Tina4 Node.js is written in TypeScript. Full type safety. Autocompletion in every editor. Compile-time error checking. Bugs surface before the server starts, not after.

WebSocket native. Node.js handles WebSocket connections without additional extensions. Real-time features -- chat, live dashboards, push notifications -- work out of the box. No Swoole. No extra gems. The runtime supports it.

Watch Out For

Async complexity. Modern async/await syntax is clean. The underlying tooling is not always clean. Unhandled promise rejections, callback-style legacy APIs, event loop blocking -- these traps exist. If your team is new to async JavaScript, expect a learning curve. The syntax is simple. The debugging is not.

Single-threaded for user code. Node.js runs your code on one thread. CPU-heavy operations block the event loop. Image processing, PDF generation, complex calculations -- offload these to worker threads or queue them for background processing. The event loop handles I/O brilliantly. It handles computation poorly.

Dependency gravity. The Node.js ecosystem gravitates toward micro-dependencies. Tina4 itself has zero core dependencies. Your application's other packages might pull in hundreds of transitive dependencies. Be deliberate about what enters your `node_modules`. Every `npm install` is a decision. Treat it as one.

Quick Start

```
npm install tina4-nodejs
tina4 init nodejs my-project
cd my-project
tina4 serve
```

Comparison Table

Factor	Python	PHP	Ruby	Node.js
Install	`pip install tina4-python`	`composer require tina4stack/tina4php`	`gem install tina4ruby`	`npm install tina4-nodejs`
CLI	`tina4`	`tina4`	`tina4`	`tina4`
Naming	`snake_case`	`camelCase`	`snake_case`	`camelCase`
Async	Native (asyncio)	Swoole extension	Rack-based	Native (event loop)
Hosting	VPS, containers, PaaS	Everywhere	VPS, containers, PaaS	VPS, containers, PaaS
Raw speed	Good	Very good (JIT)	Adequate	Best
Test count	2,112	2,220	2,333	2,646
Best for	Data/ML teams	Web agencies, existing PHP	Startups, clean code	JS/TS full-stack teams
WebSocket	Native async	Swoole required	Rack hijack / Puma	Native
Routing style	Decorator-based	Static method calls	DSL blocks	File-based + decorators
Learning curve	Low (if you know Python)	Low (if you know PHP)	Low (if you know Ruby)	Low (if you know JS/TS)
Framework LOC	~26,000	~35,000	~24,000	~32,000
Docker image	~60MB	~50MB	~70MB	~40MB
Community size	Largest (general)	Largest (web-specific)	Smallest	Large
Shared hosting	Rare	Universal	Rare	Rare
Package manager	pip	Composer	RubyGems	npm

Switching Languages

All four implementations share the same skeleton. Switching languages is not a rewrite. It is a translation.

Five things transfer without any changes:

- `src/templates/` -- Frond syntax is identical. Every template works on every backend.
- `.env` -- Same variables. Same defaults. Same priority chain.
- `src/migrations/` -- Same SQL. The database does not care what language talks to it.
- `src/public/` -- Same static files. CSS, JavaScript, images. Untouched.
- `frond.js` frontend code -- Same API. The backend language is invisible to the browser.

Three things need translation:

- `src/routes/` -- Route handlers rewritten in the new language's syntax.
- `src/orm/` -- Model definitions rewritten in the new language's class system.
- `tests/` -- Test files rewritten for the new language's test framework.

The framework knowledge transfers completely. The project structure stays. The templates stay. The migrations stay. The configuration stays. You translate your business logic. Nothing else.

How to Decide

Five questions. Answer them in order. Stop at the first clear answer.

- **What does your team already know?** Use that language. The Tina4 learning curve is the same across all four. The learning curve for a new programming language is not. Do not add two learning curves when one will do.
- **Where are you deploying?** Shared hosting means PHP. Containers mean anything. Serverless platforms -- check which runtimes are supported before you start writing code.
- **What sits next to the web app?** ML models mean Python. A React or Vue frontend team means Node.js. An existing PHP codebase means PHP. The web application is rarely the only software in the system. Choose the language that fits the neighborhood.
- **How many concurrent connections do you need?** Chat applications, live dashboards, thousands of WebSocket connections -- use Python or Node.js for native async. PHP needs Swoole. Ruby handles moderate concurrency with Puma. Know your ceiling before you build.
- **Does raw performance matter?** For most applications, all four are fast enough. The framework overhead is sub-millisecond in every language. The bottleneck is your database queries, not your router. But if you need 10,000+ requests per second on a single instance, benchmark with your actual workload. Numbers on paper are not numbers in production.

If none of these questions produce a clear answer, use Python. It is the reference implementation. It has the largest general-purpose community. It is the most-taught language in the world. When all else is equal, start where the most help exists.

Environment Variables

Every piece of Tina4 configuration lives in one file. A `.env` at the root of your project. All optional. All with sensible defaults. Identical across Python, PHP, Ruby, and Node.js. This chapter is the complete reference.

How .env Files Work

A `.env` file is plain text. Key-value pairs. Nothing more.

```
# This is a comment
TINA4_DATABASE_URL=sqlite:///data/app.db
TINA4_DEBUG=true

# Blank lines are ignored

# Values with spaces need quotes
TINA4_MAIL_FROM="My App <noreply@example.com>"

# No quotes needed for simple values
TINA4_SECRET=my-secret-key-change-in-production
```

Rules

Eight rules govern the format, and all four frameworks follow every one of them:

- One variable per line.
- Format is `KEY=VALUE`. Whitespace around the key is trimmed.
- Lines starting with `#` are comments.
- Blank lines are ignored.
- A shell-style `export KEY=value` works. Paste straight from a shell profile.
- An unquoted value ends at the first `#`, so `KEY=value # note` sets `value`.
- Quotes decide how the value is read:
- Double quotes (`"value"`) strip the quotes, process `\n`, `\t` and `\\`, and expand `${VAR}` references. A `#` inside stays a `#`.
- Single quotes (`'value'`) strip the quotes and change nothing else. No escapes, no expansion. This is how you keep a literal `${...}`.
- `${VAR}` expands against variables already loaded plus the real environment. A name that resolves to nothing is left exactly as written and warned about, so a typo is visible instead of silently becoming an empty string. A bare `$OTHER_VAR` without braces is always a literal.

```
HOST=example.com
API=${HOST}/api          # becomes example.com/api
LITERAL='${HOST}/api'    # stays ${HOST}/api
```

A line the parser cannot use, one with no `=` or an invalid key, is skipped with a warning naming the line number. It never fails in silence, and one bad line never stops the rest of the file loading.

Simple format. No surprises. A developer who has never seen a `.env` file understands it in thirty seconds.

The .env File Is Not Committed to Git

The `.env` file holds secrets. Database passwords. JWT keys. API tokens. It belongs in `.gitignore`. When you run `tina4 init`, the generated `.gitignore` already excludes it.

Commit a `.env.example` instead. Placeholder values. A map for the next developer:

```
# .env.example -- copy to .env and fill in real values
TINA4_DATABASE_URL=sqlite:///data/app.db
TINA4_DEBUG=false
TINA4_SECRET=CHANGE_ME
TINA4_MAIL_HOST=smtp.example.com
TINA4_MAIL_USERNAME=
TINA4_MAIL_PASSWORD=
```

The example file documents what the application expects. The real file stays on the machine that runs it. Never in the repository.

The Priority Chain

Tina4 resolves every configurable value through a three-level chain:

```
Constructor argument  >  .env file  >  Hardcoded default
```

Three levels. Strict order. No exceptions.

- **Constructor argument wins.** Pass a value in code and it overrides everything.
- **.env file is second.** No code override? The `.env` value takes over.
- **Default is last.** Neither code nor `.env` specifies a value? The framework's built-in default applies.

This pattern holds across all variables and all four language implementations. Learn it once. Apply it everywhere.

Boolean Values

Environment variables are strings. The `.env` file has no concept of `true` or `false`. Tina4 recognises these values as truthy:

Value	Treated as
`true`, `True`, `TRUE`	`true`
`1`	`true`
`yes`, `Yes`, `YES`	`true`
`on`, `On`, `ON`	`true`

Everything else is `false`:

Value	Treated as
`false`, `0`, `no`, `off`	`false`
(empty string)	`false`
(variable not set)	`false`

Write whichever style your team prefers:

```
# All of these enable debug mode
TINA4_DEBUG=true
TINA4_DEBUG=1
TINA4_DEBUG=yes
```

Complete .env Reference

Server

Variable	Default	Description
`HOST`	`0.0.0.0`	Bind address. `0.0.0.0` listens on all interfaces (required for Docker). `127.0.0.1` restricts to localhost.
`PORT`	See below	HTTP server port. Each framework has a unique default to avoid conflicts when running side-by-side.
`TINA4_DEBUG`	`false`	Master toggle. Enables debug overlay, full stack traces, Swagger UI, live reload, query logging. **Never `true` in production.**

Default ports by framework:

Framework	Default Port
PHP	`7145`
Python	`7146`
Ruby	`7147`
Node.js	`7148`

Authentication

Variable	Default	Description
`TINA4_SECRET`	`tina4-default-secret`	Secret key for JWT signing (HMAC-SHA256). Long, random, never committed to git. **Change this in production.**
`TINA4_API_KEY`	*(none)*	Static API key for bearer token authentication. When set, requests with `Authorization: Bearer {TINA4_API_KEY}` are accepted.
`TINA4_TOKEN_LIMIT`	`60`	Token lifetime in minutes. Tokens issued by `get_token()` / `getToken()` expire after this many minutes.

CSRF Protection

Variable	Default	Description
`TINA4_CSRF`	`true`	Enable CSRF token validation on POST/PUT/PATCH/DELETE. Set to `false` to disable (e.g. for internal microservices behind a firewall).

CSRF is **on by default**. When enabled:

- POST/PUT/PATCH/DELETE requests must include a `formToken` in the request body or an `X-Form-Token` header.
- GET/HEAD/OPTIONS requests are not checked.
- Requests with a valid `Authorization: Bearer` token skip CSRF validation.
- Routes marked `@noauth()` skip CSRF validation.
- Tokens in query strings are **rejected** (security risk).

To disable for internal services:

```
TINA4_CSRF=false
```

Database

Variable	Default	Description
`TINA4_DATABASE_URL`	`sqlite:///data/app.db`	Connection string. The URL scheme selects the driver.
`TINA4_DATABASE_USERNAME`	*(from URL)*	Override the username in `TINA4_DATABASE_URL`. Useful when credentials contain special characters.
`TINA4_DATABASE_PASSWORD`	*(from URL)*	Override the password in `TINA4_DATABASE_URL`.
`TINA4_AUTOCOMMIT`	`true`	Standalone writes auto-commit on their own connection (durable + visible across a pool); explicit transactions stay atomic. Set `false` for strict manual-commit mode.

Connection string formats:

```
# SQLite (default -- no credentials needed)
TINA4_DATABASE_URL=sqlite:///data/app.db

# PostgreSQL
TINA4_DATABASE_URL=postgresql://user:password@hostname:5432/database_name

# MySQL / MariaDB
TINA4_DATABASE_URL=mysql://user:password@hostname:3306/database_name

# Microsoft SQL Server
TINA4_DATABASE_URL=mssql://user:password@hostname:1433/database_name

# Firebird
TINA4_DATABASE_URL=firebird://user:password@hostname:3050/path/to/database.fdb

# MongoDB (SQL queries are auto-translated)
TINA4_DATABASE_URL=mongodb://user:password@hostname:27017/database_name
```

Gotcha: Special characters in your database password -- `@`, `#`, `:`, `/` -- will break URL parsing. URL-encode them (`@` becomes `%40`) or split the credentials into separate variables.

DB Query Cache

Variable	Default	Description
`TINA4_DB_CACHE`	`false`	Enable in-memory caching of query results.
`TINA4_DB_CACHE_TTL`	`30`	Cache time-to-live in seconds.

ORM

Variable	Default	Description
`TINA4_ORM_PLURAL_TABLE_NAMES`	`false`	Append "s" to auto-generated table names. When `false` (default), `Product` maps to `product`. When `true`, `Product` maps to `products`.

CORS (Cross-Origin Resource Sharing)

Variable	Default	Description
`TINA4_CORS_ORIGINS`	*(empty)*	Comma-separated allowed origins. Empty denies every cross-origin request; `*` allows all and has to be set explicitly. In production, list your actual domains.
`TINA4_CORS_METHODS`	`GET,POST,PUT,PATCH,DELETE,OPTIONS`	Comma-separated HTTP methods allowed in cross-origin requests.
`TINA4_CORS_HEADERS`	`Content-Type,Authorization,X-Request-ID`	Comma-separated headers the client is allowed to send.
`TINA4_CORS_CREDENTIALS`	`true`	Whether the browser sends cookies and auth headers in cross-origin requests.
`TINA4_CORS_MAX_AGE`	`86400`	How long (seconds) the browser caches preflight responses. `86400` = 24 hours.

Gotcha: `TINA4_CORS_ORIGINS=*` combined with `TINA4_CORS_CREDENTIALS=true` is invalid per the CORS spec. Tina4 handles this automatically -- when origin is `*`, the credentials header is not sent.

Security Headers

Variable	Default	Description
`TINA4_FRAME_OPTIONS`	`SAMEORIGIN`	`X-Frame-Options` header. Prevents clickjacking. Options: `DENY`, `SAMEORIGIN`.
`TINA4_HSTS`	*(empty/off)*	`Strict-Transport-Security` max-age in seconds. Set to `31536000` (1 year) in production with HTTPS.
`TINA4_CSP`	`default-src 'self'`	`Content-Security-Policy` header. Controls which resources the browser is allowed to load.
`TINA4_REFERRER_POLICY`	`strict-origin-when-cross-origin`	`Referrer-Policy` header. Controls what referrer info is sent with requests.
`TINA4_PERMISSIONS_POLICY`	`camera=(), microphone=(), geolocation=()`	`Permissions-Policy` header. Disables browser features your app doesn't need.

The `X-Content-Type-Options: nosniff` and `X-XSS-Protection: 0` headers are always set (no env variable -- these are security best practices).

Rate Limiter

Variable	Default	Description
`TINA4_RATE_LIMIT`	`100`	Maximum requests per window per IP address.
`TINA4_RATE_WINDOW`	`60`	Window duration in seconds. Default: 100 requests per 60 seconds.

The rate limiter adds three headers to every response:

```
X-RateLimit-Limit: 100
X-RateLimit-Remaining: 85
X-RateLimit-Reset: 60
```

When the limit is exceeded, the server returns `429 Too Many Requests`.

Sessions

Variable	Default	Description
`TINA4_SESSION_BACKEND`	`file`	Session storage backend. Options: `file`, `redis`, `valkey`, `mongo`, `database`.
`TINA4_SESSION_PATH`	`data/sessions`	Directory for file-based sessions. Relative to the project root.
`TINA4_SESSION_TTL`	`3600`	Session expiry in seconds. Default: 1 hour.
`TINA4_SESSION_SAMESITE`	`Lax`	SameSite cookie attribute. Options: `Strict`, `Lax`, `None`.

Redis/Valkey Session Backend

Variable	Default	Description
`TINA4_SESSION_REDIS_HOST`	`127.0.0.1`	Redis host.
`TINA4_SESSION_REDIS_PORT`	`6379`	Redis port.
`TINA4_SESSION_REDIS_PASSWORD`	*(none)*	Redis password.
`TINA4_SESSION_REDIS_DB`	`0`	Redis database number.
`TINA4_SESSION_REDIS_PREFIX`	`tina4:session:`	Key prefix for session data.
`TINA4_SESSION_VALKEY_HOST`	`localhost`	Valkey host.
`TINA4_SESSION_VALKEY_PORT`	`6379`	Valkey port.
`TINA4_SESSION_VALKEY_PASSWORD`	*(none)*	Valkey password.
`TINA4_SESSION_VALKEY_DB`	`0`	Valkey database number.
`TINA4_SESSION_VALKEY_PREFIX`	`tina4:session:`	Key prefix for session data.

MongoDB Session Backend

Variable	Default	Description
`TINA4_SESSION_MONGO_URI`	*(none)*	Full MongoDB connection URI. Overrides host/port.
`TINA4_SESSION_MONGO_HOST`	`localhost`	MongoDB host.
`TINA4_SESSION_MONGO_PORT`	`27017`	MongoDB port.
`TINA4_SESSION_MONGO_USERNAME`	*(none)*	MongoDB username.
`TINA4_SESSION_MONGO_PASSWORD`	*(none)*	MongoDB password.
`TINA4_SESSION_MONGO_DB`	`tina4_sessions`	MongoDB database name.
`TINA4_SESSION_MONGO_COLLECTION`	`sessions`	MongoDB collection for session data.

Queue

Variable	Default	Description
`TINA4_QUEUE_BACKEND`	`file`	Queue storage backend. Options: `file`, `rabbitmq`, `kafka`, `mongodb`.
`TINA4_QUEUE_PATH`	`data/queue`	Directory for file-based queue storage (when using `file` backend).
`TINA4_QUEUE_URL`	*(none)*	Generic connection URL for queue backend.

RabbitMQ Queue Backend

Variable	Default	Description
`TINA4_RABBITMQ_HOST`	`localhost`	RabbitMQ host.
`TINA4_RABBITMQ_PORT`	`5672`	RabbitMQ port.
`TINA4_RABBITMQ_USERNAME`	`guest`	RabbitMQ username.
`TINA4_RABBITMQ_PASSWORD`	`guest`	RabbitMQ password.
`TINA4_RABBITMQ_VHOST`	`/`	RabbitMQ virtual host.

Kafka Queue Backend

Variable	Default	Description
`TINA4_KAFKA_BROKERS`	`localhost:9092`	Comma-separated Kafka broker addresses.
`TINA4_KAFKA_GROUP_ID`	`tina4_consumer_group`	Kafka consumer group ID.

MongoDB Queue Backend

Variable	Default	Description
`TINA4_MONGO_URI`	*(none)*	Full MongoDB connection URI. Overrides host/port.
`TINA4_MONGO_HOST`	`localhost`	MongoDB host.
`TINA4_MONGO_PORT`	`27017`	MongoDB port.
`TINA4_MONGO_USERNAME`	*(none)*	MongoDB username.
`TINA4_MONGO_PASSWORD`	*(none)*	MongoDB password.
`TINA4_MONGO_DB`	`tina4`	MongoDB database name.
`TINA4_MONGO_COLLECTION`	`tina4_queue`	MongoDB collection for queue messages.

Response Cache

Variable	Default	Description
`TINA4_CACHE_BACKEND`	`memory`	Cache storage. Options: `memory` (in-process), `file`, `redis`, `valkey`, `memcached`, `mongodb`, `database`. Falls back to `file` if the configured backend is unreachable.
`TINA4_CACHE_TTL`	`60`	Default cache TTL in seconds for cached routes.
`TINA4_CACHE_MAX_ENTRIES`	`1000`	Maximum cached responses (memory backend). Oldest entries evicted at the limit.
`TINA4_CACHE_DIR`	`data/cache`	Directory for file-based cache.
`TINA4_CACHE_URL`	`redis://localhost:6379`	Connection URL for remote cache backends (Redis, Valkey, Memcached, MongoDB). For `database`, falls back to `TINA4_DATABASE_URL`.
`TINA4_CACHE_USERNAME`	*(none)*	Username for the cache backend. May also be embedded in `TINA4_CACHE_URL`.
`TINA4_CACHE_PASSWORD`	*(none)*	Password for the cache backend. May also be embedded in `TINA4_CACHE_URL` (e.g. `redis://:pass@host`). Memcached is unauthenticated.

Logging

Variable	Default	Description
`TINA4_LOG_LEVEL`	`ERROR`	Minimum log level. Options: `ALL`, `DEBUG`, `INFO`, `WARNING`, `ERROR`.
`TINA4_LOG_MAX_SIZE`	`10`	Maximum log file size in MB before rotation.
`TINA4_LOG_KEEP`	`5`	Number of rotated log files to keep.

Messenger (Email / SMTP)

Tina4 supports two naming conventions for SMTP variables. The `SMTP_*` variables are the primary names. The `TINA4_MAIL_*` variables are aliases that take precedence when both are set.

Variable	Alias	Default	Description
`TINA4_MAIL_HOST`	`TINA4_MAIL_HOST`	`localhost`	SMTP server hostname.
`TINA4_MAIL_PORT`	`TINA4_MAIL_PORT`	`587`	SMTP port. `587` (TLS), `465` (SSL), `25` (unencrypted).
`TINA4_MAIL_USERNAME`	`TINA4_MAIL_USERNAME`	*(none)*	SMTP authentication username.
`TINA4_MAIL_PASSWORD`	`TINA4_MAIL_PASSWORD`	*(none)*	SMTP authentication password.
`TINA4_MAIL_FROM`	`TINA4_MAIL_FROM`	`noreply@localhost`	Default sender address.
`TINA4_MAIL_FROM_NAME`	`TINA4_MAIL_FROM_NAME`	*(none)*	Sender display name.
`TINA4_MAIL_ENCRYPTION`	-	`tls`	Connection encryption. `tls`, `ssl`, or `none`.
`TINA4_MAIL_CAPTURE`	-	`false`	Force local DevMailbox capture and skip SMTP, even when a host exists.

`TINA4_MAIL_REDIRECT_TO`	-	*(none)*	Replace every SMTP recipient with a comma-separated safety list.

IMAP (for reading email)

Variable	Default	Description
`TINA4_MAIL_IMAP_HOST`	*(falls back to TINA4MAILHOST)*	IMAP server hostname.
`TINA4_MAIL_IMAP_PORT`	`993`	IMAP port (993 = SSL).

Localization (i18n)

Variable	Default	Description
`TINA4_LOCALE`	`en`	Default locale. Determines which translation file (`src/locales/{locale}.json`) is loaded.
`TINA4_LOCALE_DIR`	`src/locales`	Directory containing translation JSON files.

Swagger / OpenAPI

Variable	Default	Description
`TINA4_SWAGGER_TITLE`	`Tina4 API`	API title shown in Swagger UI.
`TINA4_SWAGGER_VERSION`	`1.0.0`	API version shown in Swagger UI.
`TINA4_SWAGGER_DESCRIPTION`	*(none)*	API description.

File Uploads

Variable	Default	Description
`TINA4_MAX_UPLOAD_SIZE`	`10485760`	Maximum upload size in bytes. Default: 10 MB.

WebSocket

Variable	Default	Description
`TINA4_WS_PORT`	`8080`	WebSocket server port (when running as separate process).
`TINA4_WS_BACKPLANE`	*(none)*	WebSocket backplane type. Set to `redis` to relay broadcasts across instances.
`TINA4_WS_BACKPLANE_URL`	`redis://localhost:6379`	Connection URL for the WebSocket backplane.

Services (Background Workers)

Variable	Default	Description
`TINA4_SERVICE_DIR`	`src/services`	Directory for service worker scripts.
`TINA4_SERVICE_SLEEP`	`5`	Seconds between service worker ticks.

Dev Mailbox

Variable	Default	Description
`TINA4_MAILBOX_DIR`	`data/mailbox`	Directory for captured mail. It does not control whether capture runs.

`TINA4_DEBUG` exposes the mailbox in the developer dashboard but does not stop SMTP. On staging, choose one explicit safety policy:

```
# Keep all mail inside the application.
TINA4_MAIL_CAPTURE=true

# Or exercise SMTP while replacing every original To, Cc, and Bcc recipient.
TINA4_MAIL_REDIRECT_TO=qa@example.com,product@example.com
```

Capture wins if both settings exist. Redirect applies only to real SMTP delivery and records the original recipients in `X-Tina4-Original-To`.

Minimal .env for Development

Getting started? One line:

```
TINA4_DEBUG=true
```

Everything else uses sensible defaults:

- Binds to `0.0.0.0` on the framework's default port
- SQLite database at `data/app.db`
- File-based sessions (1 hour TTL)
- File-based queue
- In-memory response cache
- CORS allows all origins
- 100 requests per minute rate limit
- CSRF protection enabled
- Security headers active

One line. A working development environment. Add variables when you need them. Not before.

Minimal .env for Production

```
TINA4_DEBUG=false
TINA4_SECRET=a-very-long-random-string-at-least-32-characters
TINA4_DATABASE_URL=postgresql://app_user:strong_password@db-host:5432/myapp
TINA4_CORS_ORIGINS=https://myapp.com
TINA4_HSTS=31536000
TINA4_MAIL_HOST=smtp.sendgrid.net
TINA4_MAIL_PORT=587
TINA4_MAIL_USERNAME=apikey
TINA4_MAIL_PASSWORD=SG.xxxxx
TINA4_MAIL_FROM=noreply@myapp.com
```

Ten lines. A production application. Debug disabled. Real database. Signed tokens. Locked CORS. HSTS enabled. Email configured. Everything else keeps its defaults.

Docker .env

When running in Docker, `HOST` must be `0.0.0.0` so the container accepts connections from outside. This is already the default, but if you override it, keep this in mind:

```
# Required for Docker -- do NOT set to 127.0.0.1
HOST=0.0.0.0
PORT=7145
TINA4_DEBUG=false
```

Full .env Template

Copy this to your `.env.example` as a starting point:

```
# ==============================================================================
# Tina4 Environment Configuration
# Copy this file to .env and fill in your values
# ==============================================================================

# --- Server ---
HOST=0.0.0.0
```

```
# PORT=7145
TINA4_DEBUG=false

# --- Authentication ---
TINA4_SECRET=CHANGE_ME
# TINA4_API_KEY=
TINA4_TOKEN_LIMIT=60

# --- CSRF ---
TINA4_CSRF=true

# --- Database ---
TINA4_DATABASE_URL=sqlite:///data/app.db
# TINA4_DATABASE_USERNAME=
# TINA4_DATABASE_PASSWORD=
# TINA4_AUTOCOMMIT=false

# --- DB Query Cache ---
# TINA4_DB_CACHE=false
# TINA4_DB_CACHE_TTL=30

# --- ORM ---
# TINA4_ORM_PLURAL_TABLE_NAMES=false

# --- CORS ---
TINA4_CORS_ORIGINS=*
TINA4_CORS_METHODS=GET,POST,PUT,PATCH,DELETE,OPTIONS
TINA4_CORS_HEADERS=Content-Type,Authorization,X-Request-ID
TINA4_CORS_CREDENTIALS=true
TINA4_CORS_MAX_AGE=86400

# --- Security Headers ---
# TINA4_FRAME_OPTIONS=SAMEORIGIN
# TINA4_HSTS=
# TINA4_CSP=default-src 'self'
# TINA4_REFERRER_POLICY=strict-origin-when-cross-origin
# TINA4_PERMISSIONS_POLICY=camera=(), microphone=(), geolocation=()

# --- Rate Limiting ---
TINA4_RATE_LIMIT=100
TINA4_RATE_WINDOW=60

# --- Logging ---
TINA4_LOG_LEVEL=ERROR
# TINA4_LOG_MAX_SIZE=10
# TINA4_LOG_KEEP=5

# --- Sessions ---
TINA4_SESSION_BACKEND=file
# TINA4_SESSION_PATH=data/sessions
TINA4_SESSION_TTL=3600
# TINA4_SESSION_REDIS_HOST=127.0.0.1
# TINA4_SESSION_REDIS_PORT=6379
# TINA4_SESSION_VALKEY_HOST=localhost
# TINA4_SESSION_VALKEY_PORT=6379

# --- Queue ---
TINA4_QUEUE_BACKEND=file
# TINA4_RABBITMQ_HOST=localhost
```

```
# TINA4_RABBITMQ_PORT=5672
# TINA4_RABBITMQ_USERNAME=guest
# TINA4_RABBITMQ_PASSWORD=guest
# TINA4_KAFKA_BROKERS=localhost:9092
# TINA4_KAFKA_GROUP_ID=tina4_consumer_group
# TINA4_MONGO_HOST=localhost
# TINA4_MONGO_PORT=27017
# TINA4_MONGO_DB=tina4
# TINA4_MONGO_COLLECTION=tina4_queue

# --- Response Cache ---
TINA4_CACHE_BACKEND=memory
TINA4_CACHE_TTL=60
TINA4_CACHE_MAX_ENTRIES=1000

# --- Email (SMTP) ---
# TINA4_MAIL_HOST=smtp.example.com
# TINA4_MAIL_PORT=587
# TINA4_MAIL_USERNAME=
# TINA4_MAIL_PASSWORD=
# TINA4_MAIL_FROM=noreply@example.com

# --- Localization ---
TINA4_LOCALE=en

# --- Swagger ---
TINA4_SWAGGER_TITLE=Tina4 API
TINA4_SWAGGER_VERSION=1.0.0
# TINA4_SWAGGER_DESCRIPTION=

# --- File Uploads ---
# TINA4_MAX_UPLOAD_SIZE=10485760

# --- Services ---
# TINA4_SERVICE_DIR=src/services
# TINA4_SERVICE_SLEEP=5
```

Summary

Count	Category
3	Server (HOST, PORT, TINA4_DEBUG)
3	Authentication (TINA4*SECRET, TINA4*API*KEY, TINA4*TOKEN_LIMIT)
1	CSRF (TINA4_CSRF)
4	Database (TINA4*DATABASE*URL, USERNAME, PASSWORD, AUTOCOMMIT)
2	DB query cache
5	CORS
5	Security headers
2	Rate limiter
3	Logging
14	Sessions (base + Redis + Valkey + MongoDB)
15	Queue (base + RabbitMQ + Kafka + MongoDB)
5	Response cache
10	Messenger (SMTP + IMAP + staging safety)
2	Localization
3	Swagger
1	File uploads
3	WebSocket
2	Services
1	Dev mailbox
84	**Total**

Every variable follows the same priority chain: constructor > `.env` > default. Every boolean is interpreted consistently across all four frameworks. Every variable has a sensible default that works for development without any configuration.

One file. Eighty-four knobs. Turn what you need. Leave the rest alone.

DevOps and Deployment

Someone handed you a Tina4 application and asked you to run it. You have never seen the framework before. This page tells you what you need and nothing else.

Tina4 apps have no runtime dependencies. No package tree to resolve, no native addons to compile, no build step that fails on a fresh machine. The application directory and a language runtime are the whole story. That makes deployment short, and it makes this chapter short with it.

1. Know Which Server You Are Running

Every Tina4 app answers HTTP, but the thing underneath differs by language. That one fact decides most of your deployment.

Language	HTTP server	Concurrency
Python	asyncio, plus an ASGI entry point	One event loop, or workers under uvicorn / gunicorn / granian
PHP	Tina4's own accept loop, php-fpm, or openswoole	Your choice, see below
Ruby	WEBrick	Threads
Node.js	Native `node:http`	One event loop

Python, Ruby and Node have one sensible answer each, and `tina4 deploy docker` generates it:

```
tina4 deploy docker
```

The image starts the app with `tina4 serve --production`. Nothing else to pick.

Python has one extra door worth knowing about. The framework exposes an ASGI entry point, so you can run it under uvicorn, hypercorn or granian and use their worker models instead of the built-in loop. Reach for that when one event loop stops being enough.

2. PHP Only: Pick a Runtime

PHP is the one language where the process model is your decision, because it has three real answers rather than one.

Runtime	What serves the request	Pick it when
Built-in server	Tina4's own accept loop	You want one process, no front end, nothing to install
php-fpm + nginx	A php-fpm worker per request	You want the boring option your ops team already knows

openswoole	A resident Swoole worker	Per-request bootstrap is your bottleneck and you will manage resident state

```
tina4 deploy docker                    # built-in server (default)
tina4 deploy docker --runtime fpm      # nginx + php-fpm
tina4 deploy docker --runtime swoole   # openswoole
```

Each writes a Dockerfile and the files that Dockerfile needs. The swoole image adds `server.php`. The fpm image adds `nginx.fpm.conf` and `docker-entrypoint.fpm.sh`. Read them, commit them, treat them as yours.

The default image installs `pcntl` and forks a process per request out of the box, verified by building and running it: twenty requests, twenty pids. Switch to the worker pool at run time without rebuilding anything:

```
docker run -e TINA4_SERVE_WORKERS=8 -p 7145:7145 your-image
```

php-fpm is the safe default

Every request gets fresh process state. Nothing leaks between requests. A fatal error in one request cannot poison the next. Your monitoring already understands it and the failure modes are twenty years old.

The cost is memory. Each worker carries its own copy of the interpreter and the application.

The built-in server needs pcntl

This is the one thing that catches people out, and it is silent.

Tina4's own PHP server handles concurrency two ways, and both need the `pcntl` extension. It forks a request so a slow handler cannot block the others, and `TINA4_SERVE_WORKERS` pre-forks a pool of long-lived workers. Without `pcntl`, both stop existing. The server still answers. Nothing logs an error. Every concurrency setting in your `.env` does nothing at all.

The stock `php:8.4-cli` image ships `posix` and not `pcntl`, so this is the normal state of a hand-built image rather than an edge case. Check it first:

```
php -m | grep -i pcntl
```

The image `tina4 deploy docker` generates installs `pcntl` and fails the build if it does not load. If you write your own Dockerfile, do the same.

Forking is the default, so there is nothing to turn on

With `pcntl` present, PHP's built-in server already forks a process per request. You do not enable it. You only lose it, and there are three ways to do that:

Cause	What you see	Fix
`pcntl` absent	The server answers from one process. No warning	Install `pcntl`

`TINA4_SERVE_WORKERS` above 1	The worker pool runs instead	Leave it at 1
`TINA4_SERVE_FORK=false`	You asked for one process	Remove the line

`TINA4_DEBUG` does not switch it off. That is deliberate. A slow route should never freeze your development server, and that is the case forking exists for.

Ask the server, do not trust the config

The failure here is silent, so check the process table rather than the settings. Add a route that reports its own pid:

```
\Tina4\Router::get("/pid", fn($rq, $rs) => $rs((string)getmypid(), 200, "text/plain"));
```

Then count how many processes answer twenty requests:

```
for i in $(seq 1 20); do curl -s http://127.0.0.1:7145/pid; echo; done | sort -u | wc -l
```

Result	Meaning
20	A process per request. Forking is live
1	One process. Work through the table above
N	The pool is running with N workers, not fork per request

Run it once after any deployment change. This check is how we found that three benchmark runs had been measuring a single-process server while every configuration file said otherwise. The settings agreed with each other and disagreed with reality, and only the process table knew.

Swoole keeps your application resident

No per-request bootstrap. The app stays in memory between requests, which is where the speed comes from and where the rules come from too.

Anything the application writes to a static or a global lives for the life of the worker, not the request. A cache nobody bounded is a leak that grows all day. Keep `TINA4_DEBUG` false, because the dev toolbar's request log is a static array that only ever grows.

If none of that appeals, the other two runtimes cannot leak between requests, because they keep nothing between requests.

The Swoole entry point

`tina4 deploy docker --runtime swoole` writes this as `server.php`. It is short because `App::__invoke()` does the work: hand it a Swoole request, get a Tina4 response back. Your routes, ORM, middleware and templates never know the difference.

```
<?php
require __DIR__ . '/vendor/autoload.php';

$app = new \Tina4\App(__DIR__);
```

```
$http = new Swoole\Http\Server(
    getenv('TINA4_SWOOLE_HOST') ?: '0.0.0.0',
    (int)(getenv('TINA4_SWOOLE_PORT') ?: 7145)
);

// OpenSwoole 22 removed the procedural swoole_*() helpers. Mainline Swoole
// still has them and has no OpenSwoole\Util. Call either one unguarded and the
// image builds, then the container exits 255 the moment it starts.
$workers = (int)(getenv('TINA4_SWOOLE_WORKERS') ?: 0);
if ($workers <= 0) {
    if (class_exists('OpenSwoole\Util')) {
        $workers = \OpenSwoole\Util::getCPUNum() * 2;
    } elseif (function_exists('swoole_cpu_num')) {
        $workers = swoole_cpu_num() * 2;
    } else {
        $workers = 4;
    }
}

$http->set([
    'worker_num'       => $workers,
    'max_request'      => (int)(getenv('TINA4_SWOOLE_MAX_REQUEST') ?: 10000),
    'enable_coroutine' => true,
]);

$http->on('request', function ($req, $res) use ($app) {
    try {
        $response = $app($req);

        $res->status($response->getStatusCode());

        foreach ($response->getHeaders() as $name => $value) {
            $res->header($name, $value);
        }

        // Cookies are a SEPARATE bag from headers. Skip this loop and every
        // Set-Cookie vanishes, which breaks sessions and logins while every
        // response still returns 200.
        foreach ($response->getCookies() as $name => $cookie) {
            $res->cookie(
                $name, $cookie['value'], $cookie['expires'], $cookie['path'],
                $cookie['domain'], $cookie['secure'], $cookie['httponly'],
                $cookie['samesite']
            );
        }

        $res->end($response->getBody());
    } catch (\Throwable $e) {
        // A throw that escapes here kills the worker and takes every in-flight
        // coroutine with it, so it is contained and logged instead.
        \Tina4\Log::error('Unhandled error in the Swoole handler: ' . $e->getMessage());
        $res->status(500);
        $res->end('Internal Server Error');
    }
});

$http->start();
```

Three details in there earn their place. `max_request` recycles a worker, which bounds a leak you have not found. The cookie loop exists because `getCookies()` is separate from `getHeaders()`, and dropping it breaks every login while every response still says 200. The try/catch exists because an escaped throw kills the worker.

`TINA4_SWOOLE_WORKERS`, `TINA4_SWOOLE_MAX_REQUEST`, `TINA4_SWOOLE_HOST` and `TINA4_SWOOLE_PORT` tune it from the environment, so the file itself rarely needs editing.

3. Node.js and Ruby: What Each One Does With a Slow Handler

PHP gets its own section because you choose its process model. Node and Ruby do not offer that choice, but they behave differently under load and the difference decides how you write handlers.

Ruby's WEBrick serves each request on its own thread. A slow handler occupies its thread and nothing else. That is the model most people expect.

Node serves every request on ONE event loop, and that changes what "slow" means. A handler that awaits gives the loop back while it waits, so it blocks nobody. A handler that computes does not, and every other request waits behind it. Measured on a Tina4 Node server, a trivial route answering while a second route was busy:

The other route was	Trivial route answered in
idle	0.032s
awaiting a 2 second timer	0.030s
running a 2 second busy loop	1.575s

The awaited route cost the trivial one nothing. The busy loop cost it a second and a half. Both took two seconds of wall clock, so you cannot spot the difference by timing the slow route itself.

This is how every single-loop runtime works, so Tina4 does not try to engineer around it. It tells you instead. When a handler holds the loop past a threshold, the server logs a warning naming the duration:

```
Event loop blocked for 1204ms. Node serves every request on one loop, so a
handler doing CPU-bound work or synchronous I/O stalls all the others for that
long. Move the work to Tina4's queue, or await it.
```

The threshold is 250ms. Change it, or turn the warning off:

```
TINA4_LOOP_LAG_WARN_MS=500   # warn at half a second instead
TINA4_LOOP_LAG_WARN_MS=0     # silence it
```

When you see that warning, you have three real fixes:

- Push the work to the queue and return straight away. Image resizing, PDF generation, report building and bulk imports all belong there.
- Replace synchronous I/O with the awaiting version. `readFileSync` blocks the loop; `await readFile` does not.

- Run more processes. `tina4 serve --production` forks one worker per CPU

core, so a blocked worker stalls only the requests routed to it. On a single-core host it stays one process, because there is nothing to spread the work across.

Development runs a single process on purpose, because hot reload, the dev dashboard and the WebSocket registry all keep per-process state. So you will see the warning in development and not in production, which is the right way round: you find the blocking handler on your own machine.

4. Environment Variables That Matter in Production

Tina4 reads its configuration from the environment. Everything below has a default that works. These are the ones worth setting on purpose.

Set these or the app tells you off

```
TINA4_DEBUG=false
TINA4_SECRET=<64 hex characters>
```

`TINA4_DEBUG=false` turns off the dev dashboard, the toolbar, the error overlay and template recompilation. Leave it true in production and you publish your stack traces.

`TINA4_SECRET` signs your JWTs. Generate it once and keep it:

```
openssl rand -hex 32
```

Leave it blank and the framework logs a warning naming the exact command above. It will not invent one for you outside local development, because a secret that regenerates on restart invalidates every token you issued.

Timeouts that stop a hang

```
TINA4_DATABASE_CONNECT_TIMEOUT=10    # seconds; 0 waits forever
TINA4_SHUTDOWN_TIMEOUT=30            # seconds to drain before force-close
```

Both work in all four languages.

`TINA4_DATABASE_CONNECT_TIMEOUT` bounds every database connect. An unreachable host used to hang the application with no error and no ceiling. When the timeout expires the message names the host, the port, the seconds elapsed and the variable, so you can act on it without reading a stack trace.

Limits that bound a hostile request, PHP only

```
TINA4_REQUEST_TIMEOUT=30             # seconds of client silence; 0 disables
TINA4_MAX_REQUEST_HEADER=65536       # bytes; answers 431 past this
TINA4_MAX_REQUEST_BODY=10485760      # bytes; answers 413 past this
```

These exist in PHP because PHP's built-in server is the only one that parses HTTP itself, on a raw socket, with no server underneath to inherit limits from. The other three sit on something that already has them: Python bounds its header read at 30 seconds and 64KB through asyncio, Ruby inherits WEBrick's limits, and Node inherits `node:http` defaults of 60 seconds for headers and 16KB per header block.

So Python, Ruby and Node are bounded. They are just not bounded by a Tina4 variable you can tune. If you need a specific ceiling on those three, set it on the server in front of them.

The PHP header cap matches what nginx and Apache allow. The body cap refuses an oversized upload on its first packet, from the declared `Content-Length`, rather than buffering all of it and then objecting.

Worker pool, PHP built-in server only

```
TINA4_SERVE_WORKERS=8                # 1 is the default: one process
TINA4_SERVE_MAX_REQUESTS=10000       # recycle a worker after N; 0 never
```

`TINA4_SERVE_WORKERS` pre-forks a pool. The parent binds the socket once, forks the workers, and supervises them. A worker that dies gets replaced, so the pool never quietly shrinks.

`TINA4_SERVE_MAX_REQUESTS` recycles a worker after it has served its quota. It bounds the damage from a leak you have not found yet. Treat it as a safety net, not a licence.

The pool refuses to start when `TINA4_DEBUG` is true, and says so in the log. The dev dashboard, hot reload and the WebSocket registry are all per-process, so a pool in development would show you one worker's traffic and reload one worker's code. That reads as a framework bug, so the framework declines.

5. Health Checks

Every Tina4 app answers a health check with no configuration:

```
curl http://localhost:7145/__health
```

```
{"status": "ok"}
```

`/health` works too and always will. It is registered as a permanent alias, so a probe written years ago keeps working when someone sets a custom path.

Point your load balancer, your Docker `HEALTHCHECK` and your Kubernetes readiness probe at `/__health`. Set `TINA4_HEALTH_PATH` if you need it somewhere else.

6. Graceful Shutdown

Send `SIGTERM` and Tina4 stops accepting first, then drains.

A connection arriving after the signal gets a clean connection refused rather than an accept followed by a reset. Requests already in flight run to completion and write their whole response. `TINA4_SHUTDOWN_TIMEOUT` bounds the drain and defaults to 30 seconds.

That default is not arbitrary. It matches Kubernetes' `terminationGracePeriodSeconds`, so the two agree out of the box:

```
spec:
  terminationGracePeriodSeconds: 30
  containers:
    - name: app
      env:
        - name: TINA4_SHUTDOWN_TIMEOUT
          value: "30"
```

Raise both together if your requests run long. Raise one alone and Kubernetes kills the pod while Tina4 is still politely draining.

7. Logging

Containers log to stdout, and Tina4 does this without being asked.

With `TINA4_LOG_OUTPUT` unset, stdout is always on and the log file is written only in development. Your production container writes no file, bloats no writable layer, and hands every line to your log driver. That is the twelve-factor behaviour and it is the default.

```
TINA4_LOG_LEVEL=INFO                # DEBUG | INFO | WARNING | ERROR | CRITICAL
TINA4_LOG_FORMAT=json               # text is the default
```

Set `TINA4_LOG_FORMAT=json` when a log aggregator is parsing the stream. Text stays the default everywhere, because a human reading `docker logs` is the more common case and JSON makes that worse.

8. A Checklist

Before the first deploy:

- `TINA4_DEBUG=false`
- `TINA4_SECRET` set to 64 hex characters from `openssl rand -hex 32`
- `php -m | grep pcntl` returns something, if you run PHP's built-in server
- Health probe points at `/__health`
- `TINA4_SHUTDOWN_TIMEOUT` matches your orchestrator's grace period
- Database credentials arrive as environment variables, not in a committed `.env`
- `TINA4_DATABASE_CONNECT_TIMEOUT` set, so a database outage fails instead of hangs
- Node deploys run `tina4 serve --production`, which forks a worker per CPU core

Eight lines. Work through them once and the application runs the way its author intended, which is the only thing anyone deploying someone else's code actually wants.